Draft version December 1, 2025

Correlation Between the X-Shape Parameters and the Bar Parameters in Boxy/Peanut Barred Galaxies

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ABSTRACT

Approximately two-thirds of disk galaxies host a bar. They play a pivotal role in the secular evolution and kinematic dynamics of the galaxy, especially in their central regions. We present a methodology to estimate the bar pattern speed directly from the peanut radius in Milky Way-like, edge-on, boxy/peanut (BP) barred galaxies. Using a set of N-body models, we simulate the evolution of a BP galaxy over 7.5 Gyr and examine the correlation between the face-on and edge-on structural properties of the BP feature. In the face-on view, we parametrize the bar amplitude $A_{\rm bar}$ and bar pattern speed Ω. We then parametrize the BP structure observed in edge-on barred galaxies using the distance of the peanut's center from the galactic center, $R_{\rm pea}$, the width of each peanut, $\sigma_{\rm pea}$, and the vertical extent of the peanut, $A_{\rm pea}$. The model demonstrates that the peanut structure is a part of the bar itself, and demonstrates significant correlations across all of these parameters. In particular, $R_{\rm pea}$ exhibits a very promising relation with both the bar pattern speed; a linear fit to the two relationships reveals a well-constrained correlation, with uncertainties tightly bounding the best-fit line. This connection is key to understanding the morphological evolution of barred galaxies.

Keywords: Barred spiral galaxies (136) — Disk galaxies (391) —Stellar Dynamics (1596) — Galaxy structure (622), Galaxy evolution (594) — N-body simulations (1083)

1. INTRODUCTION

Galactic bars are present in optical images of ∼50% of disk galaxies in the local universe (e.g. Marinova & Jogee 2007; Reese et al. 2007; Barazza et al. 2008), and recent work suggests that as many as ∼60% of presently unbarred galaxies may have hosted a bar in the past (Lu et al. 2025). Bars are long-lived and dynamically robust features, provided the host galaxy remains undisturbed. Simulation studies consistently show that bars play a pivotal role in the secular evolution of disk galaxies, especially in their central regions (e.g. Debattista et al. 2004; Kormendy & Kennicutt Jr 2004; Cheung et al. 2013). Bar formation redistributes angular momentum within the galactic disk (Athanassoula 2003) by funneling interstellar gas towards the galactic center (Piner et al. 1995; Regan & Teuben 2004). Its influence on star formation remains debated, with evidence for both enhancement and suppression (Scaloni et al. 2024).

Because of the central role bars play in shaping galaxy morphology, identifying their presence is a critical area of galaxy dynamics research. The bar pattern speed, Ω, defined as the angular frequency of bar rotation, is a fundamental dynamical parameter that governs the orbital structure and resonances of barred galaxies. Consequently, systematic measurement of Ω is essential for understanding disk evolution.

Observationally, Ω is the most challenging bar parameter to constrain, since its determination depends on particular assumptions about the underlying galaxy model. A variety of methods have been developed, including comparisons of simulated and observed gas distributions and kinematics, (see, e.g., England et al. 1990; Sempere et al. 1995; Lindblad et al. 1996; Weiner et al. 2001; Rautiainen et al. 2008; Treuthardt et al. 2008); a non-parametric method of assuming the luminosity density obeys a continuity equation, and subsequently measuring the surface brightness and velocity field in the plane of the galaxy (Tremaine & Weinberg 1984); linking morphological features to Lindblad resonances and the locations of galaxy rings (see, e.g., Buta 1986; Pérez et al. 2012); using a potential-density phase shift to determine corotation (Buta & Zhang 2009); and examining residual gas velocity fields after subtracting a model rotation curve to locate corotation (see, e.g., Canzian 1993; Sempere et al. 1995; Piñol-Ferrer et al. 2013).

Made even more difficult, is the fact that while bars are relatively easy to detect in low-inclination systems through their non-axisymmetric isophotes, highly inclined (edge-on) galaxies, they are far more difficult to identify, often requiring stellar kinematic diagnostics (Athanassoula & Bureau 1999). The most accurate and robust approach is the non-parametric method of Tremaine & Weinberg (1984), which typically constrains Ω to within $\pm 15\%$. However, it relies on integrals that vanish for perfectly plane-symmetric images, making its application unsuitable for the measurement of bar pattern speed in edge-on galaxies (Zou et al. 2019).

Instead, a prominent morphological signature of a bar in an edge-on galaxy is the presence of a boxy/peanut (BP/X) shaped bulge in the central regions (e.g. Bureau & Freeman 1999; Fragkoudi et al. 2017). These bulges are frequently observed in external disk galaxies (e.g. Laurikainen et al. 2011; Erwin & Debattista 2013; Yoshino & Yamauchi 2015). In the Milky Way, a distinct BP morphology was identified in the multiparameter model of COBE/DIRBE images of the galactic bulge (Blitz & Spergel 1991), and subsequent work has firmly established that the density profile of the central region of the Milky Way consists of a prominent boxy-peanut bulge which is part of a longer bar structure (e.g. Wegg & Gerhard 2013; Ness & Lang 2016). More recent studies demonstrate that BP/X bulges are nearly ubiquitous in galaxies with stellar mass $\gtrsim 2.5 \times 10^{10} M_{\odot}$ (Erwin & Debattista 2017).

BP/X morphology is intrinsically tied to the evolution of the stellar bar. Large-scale observational surveys have shown that the comparable fractions of barred galaxies and those with BP/X bulges support evolutionary scenarios in which such bulges arise naturally from bars (Lütticke et al. 2000). Athanassoula (2003, 2005) confirms this connection, demonstrating that stronger bars tend to produce stronger BP bulges, with the thick part of relatively weak bars appearing boxy, that of stronger bars peanut-shaped, and the very strongest exhibiting a clear X shape.

Early simulation work suggested that BP bulges arise following a brief buckling event caused by the asymmetric vertical bending of the bar. Subsequent simulations reinforced this picture, showing that BP/X structures can emerge from short-lived vertical asymmetries (e.g. Martinez-Valpuesta & Shlosman 2004; Martinez-Valpuesta et al. 2006; Łokas 2019). However, more recent evidence highlights the pivotal role of orbital resonances in the formation and amplification of these structures (Sellwood & Gerhard 2020). In this framework, "resonant sweeping" describes orbits that are heated upon crossing the vertical inner Lindblad resonance and retain that excitation even after moving out of resonance (Sellwood & Gerhard 2020; Wheeler et al. 2023; Beraldo e Silva et al. 2023).

Because bar resonances are directly tied to the bar pattern speed, studying BP/X morphology offers a valuable connection to Ω. In N-body simulations, Ω is conveniently defined as the angular rotation rate of the $m = 2$ Fourier mode of the galaxy (Debattista et al. 2020). These models provide clear perspectives in both face-on and edge-on orientations, making them especially useful for interpreting connections between BP/X structure and bar pattern speed in edge-on systems.

Several new approaches have been developed to model the BP/X structure. Smirnov & Savchenko (2020) modeled the X-shape of external galaxies by applying a Fourier distortion to the Sérsic profile (Sérsic 1968). Tahmasebzadeh et al. (2022) reconstructed the three-dimensional density of an N-body barred galaxy with a decomposition of a bulge, bar, and disk component. Wheeler et al. (2023) implemented a supermassive black hole component in their models. More recently, Dattathri et al. (2024) applied the Schwarzschild modeling code FORSTAND (Vasiliev & Valluri 2020) to edge-on barred galaxies. In this model, orbital integration is carried out in the corotating frame of the bar to preserve a time-independent gravitational potential, making Ω a critical free parameter in the fitting process. Using this method, they successfully recovered the full 3D BP/X-shaped density distribution and bar pattern speed (Dattathri et al. 2024). Building on this framework, Tahmasebzadeh et al. (2024) analyzed the orbital structure of three N-body simulations, parameterizing the BP morphology using the method of Dattathri et al. (2024), and demonstrated that the evolution of BP morphology and orbit populations is strongly tied to the angular momentum evolution of the bar. Specifically, if the bar does not slow down, the BP structure and its orbital composition remain largely unchanged.

In this work, we extend the parametrization of BP/X morphology to a broader suite of N-body simulations. We expand upon the work of Dattathri et al. (2024) to increase the efficacy and efficiency of their edge-on BP modeling, and Tahmasebzadeh et al. (2022); Tahmasebzadeh et al. (2024) by quantifying robust correlations between bar and X-shape parameters. These results provide further evidence that the peanut is an intrinsic part of the bar structure. Most notably, we introduce a new method to estimate bar pattern speed directly from the peanut radius in the edge-on galaxies, offering a practical tool for systems where kinematic measurements are inaccessible. The structure of this paper is as follows: Section 2 describes the N-body models used to simulate barred galaxy evolution; Section 3 outlines the image fitting procedure, including parametric components and key constraints; Section 4 presents the correlations between X-shape and bar parameters and demonstrates the linear scaling relation between peanut radius and bar pattern speed.

2. N-BODY SIMULATION

We utilize five N-body models of a Milky Way-like galaxy from the *Run2000* series presented in (Wheeler et al. 2023). The following initial conditions utilize setups from previous works (e.g. Debattista et al. 2017; Debattista et al. 2020; Anderson et al. 2022; Wheeler et al. 2023), and were generated via GalactICS (e.g. Kuijken & Dubinski 1995; Widrow & Dubinski 2005; Widrow et al. 2008). All models initially lack a bar and do not include a supermassive black hole. Each model consists of an exponential disk embedded in a live dark matter halo with a Navarro–Frenk–White (NFW; Navarro et al. 1996) profile, truncated at radius $r <$ 100 kpc.

The stellar disk is modeled with particles mapped onto a cylindrical polar grid nested within a larger spherical grid that contains the halo particles. Both grids share a common origin, which is periodically recentered on the disk's particle centroid to maximize spatial resolution in the dense central regions (Wheeler et al. 2023). The models are then evolved for 7.5 Gyr using the GALAXY code (Sellwood 2014). The disk is modeled with an exponential radial density profile and an isothermal vertical density profile,

$$\Sigma(R, z) = \Sigma_0 \exp(-R/R_d)\,\mathrm{sech}^2(z/z_d), \tag{1}$$

where R_d and z_d denote the radial scale length and vertical scale height, set to 2.4 kpc and 0.3 kpc respectively (Debattista et al. 2020). The disk kinematics are initialized with an exponentially decreasing velocity dispersion described by,

$$\sigma^2(R) = \sigma_{R0}^2 \exp(-R/R_\sigma), \tag{2}$$

with $R_\sigma = 2.5$ kpc and σ_{R0} as the central radial velocity dispersion. *Run2000* serves as the control model of the five N-body simulations, with a central radial velocity dispersion of $\sigma_{R0} = 128\,\mathrm{km\,s^{-1}}$, while all other models contain slight variations in σ_{R0}. The disk is made up of 6×10^6 equal-mass particles, corresponding to a total disk mass of $\simeq 5.37\times10^{10} M_\odot$.

The spherical density distribution of the halo for all five N-body models is given by

$$\rho(r) = \frac{2^{2-\gamma}\sigma_h^2}{4\pi a_h^2}\frac{C(r)}{(r/a_h)^\gamma(1+r/a_h)^{3-\gamma}}, \tag{3}$$

where σ_h sets the halo velocity dispersion, and a_h is the halo scale radius. The function $C(r)$ provides a smooth truncation of the density profile at large radii (Widrow et al. 2008), characterized by

$$C(r) = \frac{1}{2}\,\mathrm{erfc}\left(\frac{r-r_h}{\sqrt{2}\,\delta r_h}\right). \tag{4}$$

Equation (3) reduces exactly to the standard NFW distribution when $\gamma = 1$ and $r_h \to \infty$. In all simulations presented, the dark matter halo is defined by the parameters $\sigma_h = 400\,\mathrm{km\,s^{-1}}$, $a_h = 16.7\,\mathrm{kpc}$, $\gamma = 0.873$, $r_h = 100\,\mathrm{kpc}$, and $\delta r_h = 25\,\mathrm{kpc}$ (Debattista et al. 2020). The live dark matter halo contains 4×10^6 particles, each having a mass of $\simeq 1.7\times10^5 M_\odot$, totaling to a mass of $6.8\times10^{11} M_\odot$.

To determine bar parameters, we employ standard measures: bar amplitude ($A_{m=2}$) and buckling amplitude (A_buck) (Debattista et al. 2020). These parameters are derived from the $m = 2$ symmetry mode in an azimuthal Fourier expansion of the face-on disk. Both quantities are normalized by the $m = 0$ mode and defined as follows:

$$A_\mathrm{bar} = A_{m=2} = \left|\frac{\sum_k m_k e^{2i\phi_k}}{\sum_k m_k}\right| \tag{5}$$

and

$$A_\mathrm{buck} = \left|\frac{\sum_k z_k m_k e^{2i\phi_k}}{\sum_k m_k}\right|, \tag{6}$$

where m_k, ϕ_k, and z_k are the mass, azimuth, and vertical position of the kth particle, respectively. The farther A deviates from zero, the less symmetrical the galaxy is, indicating the presence of a bar.

We measure the bar pattern speed of the models using the instantaneous method of Dehnen et al. (2023), which enables a fully consistent determination of Ω from a *single* simulation snapshot. The bar pattern speed is often estimated using the traditional finite-difference method, in which the bar angle ψ is measured from two consecutive snapshots and the pattern speed is approximated as

$$\Omega(t) \approx \frac{\Delta\psi}{\Delta t}. \tag{7}$$

However, because ψ is π-periodic and the bar can rotate substantially between widely spaced outputs, this method is not viable for long intervals Δt or for singular snapshots.

Instead, the bar orientation ψ is defined as the phase of the $m = 2$ azimuthal Fourier component of the surface density within the bar region. Because ψ is a function of particle positions, and these positions change with time according to the particle velocities, its time derivative can be written as

$$\Omega \equiv \dot{\psi} = \sum_i \frac{\partial \psi}{\partial \mathbf{x}_i} \cdot \mathbf{v}_i, \tag{8}$$

allowing the instantaneous pattern speed to be evaluated directly from the snapshot's velocity field without requiring finite differencing between outputs.

A key component in obtaining an unbiased estimate of Ω is the use of a smooth radial window function $W(R)$ when selecting particles in the bar region. Sharp annular boundaries lead to large and unavoidable flux terms in $\dot{\psi}$ as stars move across the edges. Following "Equation (25)" in Dehnen et al. (2023), we adopt the maximally smooth window

$$W(R) = \frac{(1-Q)^2}{(1+2Q)}, \qquad Q = \frac{R^2 - R_m^2}{R_e^2 - R_m^2}, \tag{9}$$

where R_m is the median radius of the bar region and R_e is set equal to the inner or outer radial boundary depending on whether $R < R_m$ or $R > R_m$. This smoothly tapering function avoids divergent derivatives at the edges, ensuring that the flux term in the Fourier-based expression for $\dot{\psi}$ is properly accounted for. The resulting measurement of the bar pattern speed is therefore unbiased, internally consistent (i.e. satisfying $\psi = \int \Omega \, dt$), and robust to particle shot noise.

Fig. 1 presents the evolution of the three bar parameters discussed above: buckling amplitude ($A_{\rm buck}$), bar amplitude ($A_{m=2}$), and bar pattern speed (Ω) for the five N-body models. Each model undergoes a single buckling event. Following this, all models except *run2000* exhibit a continuous net increase in bar amplitude accompanied by a continuous net decrease in bar pattern speed. The control model, *run2000*, instead stabilizes in both amplitude and pattern speed after ~ 3 Gyr, indicating that it is has reached an equilibrium state. These results are consistent with the expected conservation of angular momentum between bar length and pattern speed, reinforcing the findings of Tahmasebzadeh et al. (2024). The observed trend arises because bars slow down as they grow, a consequence of dynamical friction exerted by the dark matter halo (Tahmasebzadeh et al. 2024).

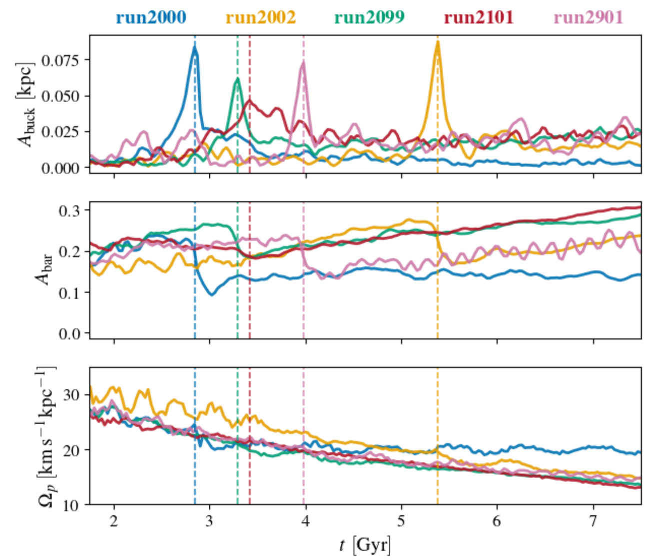

Figure 1. Evolution of bar diagnostics as a function of time for models *run2000* (blue), *run2002* (orange), *run2099* (green), *run2101* (red), and *run2901* (purple). From top to bottom the quantities displayed are: (1) the buckling amplitude ($A_{\rm buck}$), defined as the scaled $m = 2$ vertical asymmetry about the midplane; (2) the bar amplitude ($A_{m=2}$), given by the scaled $m = 2$ Fourier component of the in-plane density distribution; and (3) the bar pattern speed (Ω_p), found by the time derivative of the phase of the $m = 2$ Fourier transform. Vertical dashed lines mark the time of maximum buckling amplitude. For model *run2101*, the values of $A_{\rm buck}$ are multiplied by a factor of 1.8 for visibility.

3. METHODS

The N-body simulations allow us to sufficiently generate realistic mock photometric and kinematic data that can be analyzed by the image-fitting code IMFIT (Erwin 2015), a 2D image-fitting software for real galaxies. Specifically, we apply IMFIT to the 2D edge-on N-body snapshots to fit the X-shape and recover its structural parameters.

We use the default IMFIT minimization algorithm: a maximum-likelihood routine, employing a Levenberg-Marquardt (LM) gradient search to minimize the χ^2 statistic, defined as

$$\chi^2 = \sum_{i=1}^{N} w_i (I_{d,i} - I_{m,i})^2, \tag{10}$$

where $I_{d,i}$ and $I_{m,i}$ denote the observed and model pixel intensities, respectively. The statistical weight of each pixel, w_i, is given by

$$w_i = \frac{1}{\sigma_i^2}, \tag{11}$$

with pixel error σ_i related to intensity by $\sigma_i^2 = I_{d,i}$ under the Gaussian approximation of Poisson statistics (Erwin 2015).

We use the method of Dattathri et al. (2024) to assume a 3D parametric density distribution of the BP/X structure. The orientation of the galaxy is determined by Euler angles: where the intersection of the equatorial plane of the galaxy (x–y) and image plane (X, Y) is a line of nodes; the angle between the X axis of the image and line of nodes is the position angle θ; the angle

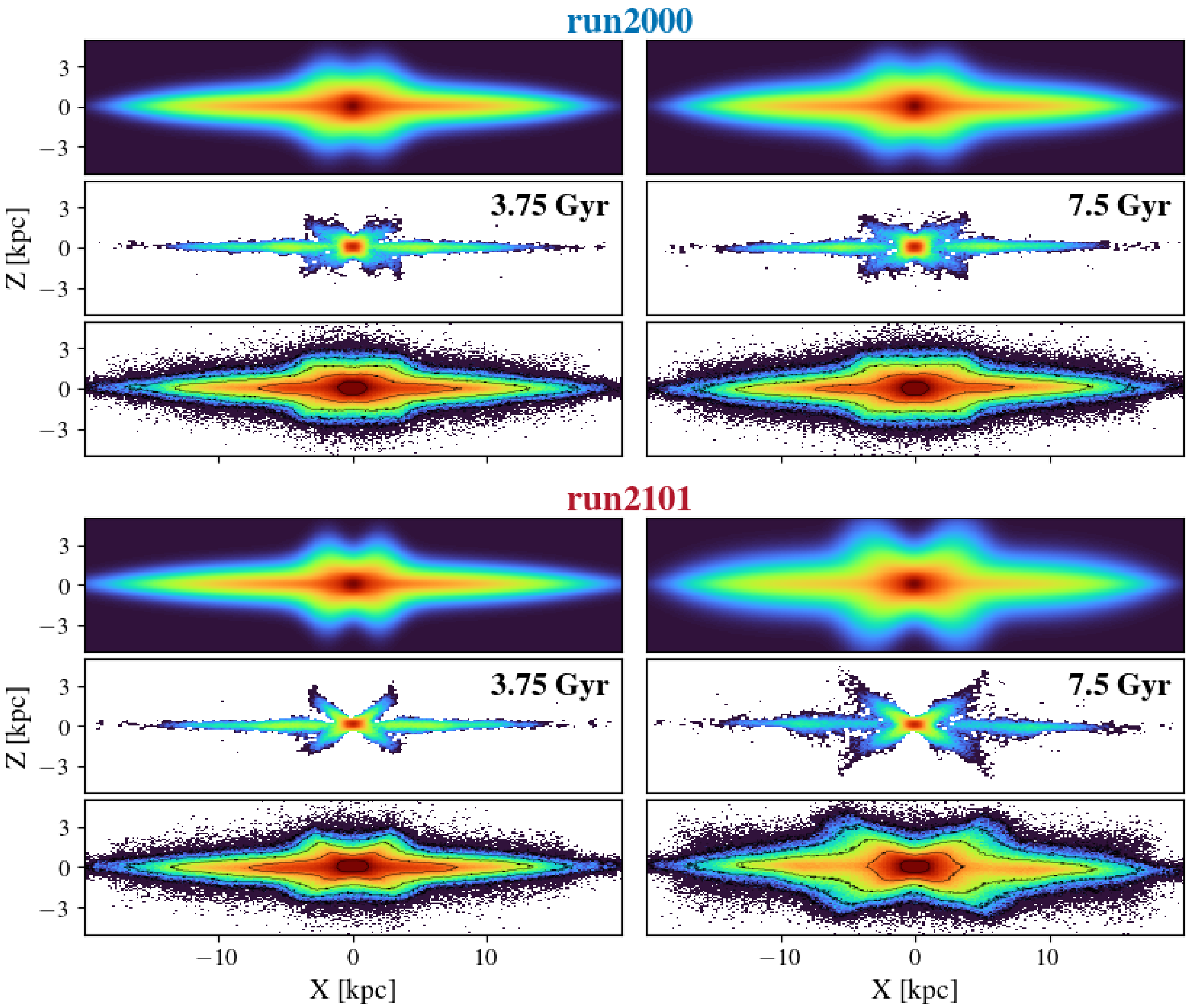

Figure 2. Edge-on structure of models *run2000* (top) and *run2101* (bottom). Columns show $t = 3.75$ and 7.5 Gyr, respectively. In each block, from top to bottom the panels display: (1) the IMFIT reconstruction of the edge-on surface density; (2) the unsharp-masked density map of the N-body model, highlighting the BP/X morphology; (3) the surface density of the N-body simulation. The left column snapshots are selected shortly after bar buckling and the formation of the BP/X structure, while the right column is taken at the end of the simulation.

between the equatorial and image plane is inclination angle i, encompassing the face-on ($i = 0°$) and edge-on ($i = 90°$) views; the angle between the bar and major axis of the galaxy (x) is ψ; and the line-of-sight (LOS) distance is s, in which $s = 0$ is the sky plane containing the galactic center. The 2D coordinate system of the sky (X, Y) is then transformed to the 3D coordinate system of the galaxy (x, y, z) by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \left[\begin{pmatrix} 1 & 0 \\ 0 & \cos i \\ 0 & \sin i \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} X - X_0 \\ Y - Y_0 \end{pmatrix} + s \begin{pmatrix} 0 \\ \sin i \\ -\cos i \end{pmatrix} \right] \times \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{12}$$

where integrating along LOS distance to $\sim \pm 5$ times the disk scale length produces the projected image of the model (Dattathri et al. 2024).

Finally, we implement the *FlatSky* function (Erwin 2015), which produces a uniform background for all pixels:

$$I(x,y) = I_{\rm sky}, \tag{13}$$

with no geometric parameters.

IMFIT offers a variety of structural components for galaxy modeling. In Section 3.1, we outline the specific set of components adopted to capture the morphology of edge-on barred galaxies with boxy/peanut bulges—implemented as a three-component bar+disk+bulge model, following the approaches of Tahmasebzadeh et al. (2021) and Dattathri et al. (2024). Section 3.2 details the parameter constraints used to initialize the fits, expanding upon the prescriptions introduced by Dattathri et al. (2024).

3.1. *Parametric Model Components*

The total parametric density distribution is constructed as the sum of three main components: bar, disk, and bulge (e.g. Tahmasebzadeh et al. 2021; Dattathri et al. 2024). Our analysis is concerned primarily with the bar component, since it is explicitly dependent on the BP/X structure. The disk and bulge components are included to ensure completeness and to obtain an accurate global fit, but they are not employed explicitly to recover BP/X parameters.

3.1.1. *Bar*

The bar is modeled using a dimensionless scaled radius R_s given by

$$\rho = \rho_0 \, \mathrm{sech}^2(-R_s), \tag{14}$$

where

$$R_s = \left\{ \left[\left(\frac{x}{X_{\rm bar}} \right)^{c_\perp} + \left(\frac{y}{Y_{\rm bar}} \right)^{c_\perp / c_\parallel} \right]^{c_\parallel} \left(\frac{z}{Z_{\rm bar}} \right)^{c_\parallel} \right\}^{1/c_\parallel}. \tag{15}$$

The coordinate system is centered at the galaxy's nucleus, with $X_{\rm bar}$, $Z_{\rm bar}$, and $Y_{\rm bar}$ denoting the semimajor-axis (x), intermediate (y), and semiminor-axis (z) lengths of the bar, respectively, in which the z-axis is perpendicular to the plane of the disk. The parameters $c_\parallel$ and $c_\perp$ regulate the diskiness/boxiness of the bar in the face-on projection, as introduced by Athanassoula et al. (1990). $c_\parallel = c_\perp = 2$ results in a pure ellipsoidal bar structure. $c_\parallel < 2$ is a disky projection, while $c_\parallel > 2$ is a boxy projection of the bar in the side-on view (Dattathri et al. 2024).

N-body simulations (Athanassoula & Misiriotis 2002) together with observational fits to the Milky Way (Wegg et al. 2015) demonstrate that the vertical scale height, $Z_{\rm bar}$, varies across the x-y plane, giving rise to the BP/X structure. This variation is modeled using a double-Gaussian distribution centered on the galactic nucleus:

$$Z_{\rm bar}(x,y) = A_{\rm pea} \exp\left(-\frac{(x - R_{\rm pea})^2}{2\sigma_{\rm pea}^2} - \frac{y^2}{2\sigma_{\rm pea}^2} \right) + A_{\rm pea} \exp\left(-\frac{(x + R_{\rm pea})^2}{2\sigma_{\rm pea}^2} - \frac{y^2}{2\sigma_{\rm pea}^2} \right) + z_0, \tag{16}$$

where the X-shape parameters, $R_{\rm pea}$ and $\sigma_{\rm pea}$, represent the distance of the peanut's center from the galactic center and the width of each peanut, respectively. The parameter $A_{\rm pea}$ quantifies the vertical extent of the peanut feature above the ellipsoidal bar's scale height, z_0. This expression closely follows the "peanut height function" described by Fragkoudi et al. (2015).

3.1.2. *Disk*

The disk is described by an axisymmetric density profile introduced by van der Kruit (1988), and enhanced by Sormani et al. (2022) through the addition of parameters:

$$\rho(R,z) = \rho_0 \exp\Big(-(R/R_{\rm disk})^k - R_{\rm hole}/R \Big) \; \mathrm{sech}(z/\alpha z_{\rm disk})^\alpha. \tag{17}$$

In this expression: $R = \sqrt{x^2 + y^2}$ denotes the radial distance in the disk plane; z measures the vertical height above the disk plane; ρ_0 is a normalization factor; $R_{\rm disc}$ is the radial scale length; $z_{\rm disc}$ is the vertical scale length; $R_{\rm hole}$ accounts for the central depletion of disk density near the corotation region; finally, α modifies the vertical profile shape (Dattathri et al. 2024).

3.1.3. *Bulge*

The center galactic bulge is represented with the Einasto (1965) profile, utilizing a triaxial generalization:

$$\rho(r) = \rho_0 \exp\left(-b_n \left(\frac{r}{R_{\rm bulge}} \right)^{1/n} - 1 \right), \tag{18}$$

where the 3D ellipsoidal radius is $r = \sqrt{x^2 + (y/q)^2 + (z/q_z)^2}$, in which from equation (15), $q = X_{\rm bar}/Y_{\rm bar}$ is the intermediate/major axis ratio, and $q_z = Z_{\rm bar}/X_{\rm bar}$ is the minor/major axis ratio. Lastly, parameter n sets the curvature of the profile, regulating how rapidly the density falls off with radius (Dattathri et al. 2024).

3.2. *Parameter Constraints*

From Dattathri et al. (2024), a bar+disk+bulge model provides a reasonable fit to Milky Way-like, edge-on, boxy/peanut-barred galaxies. In total, the model includes 30 free parameters. Such a large parameter space can easily overwhelm the LM algorithm, particularly if

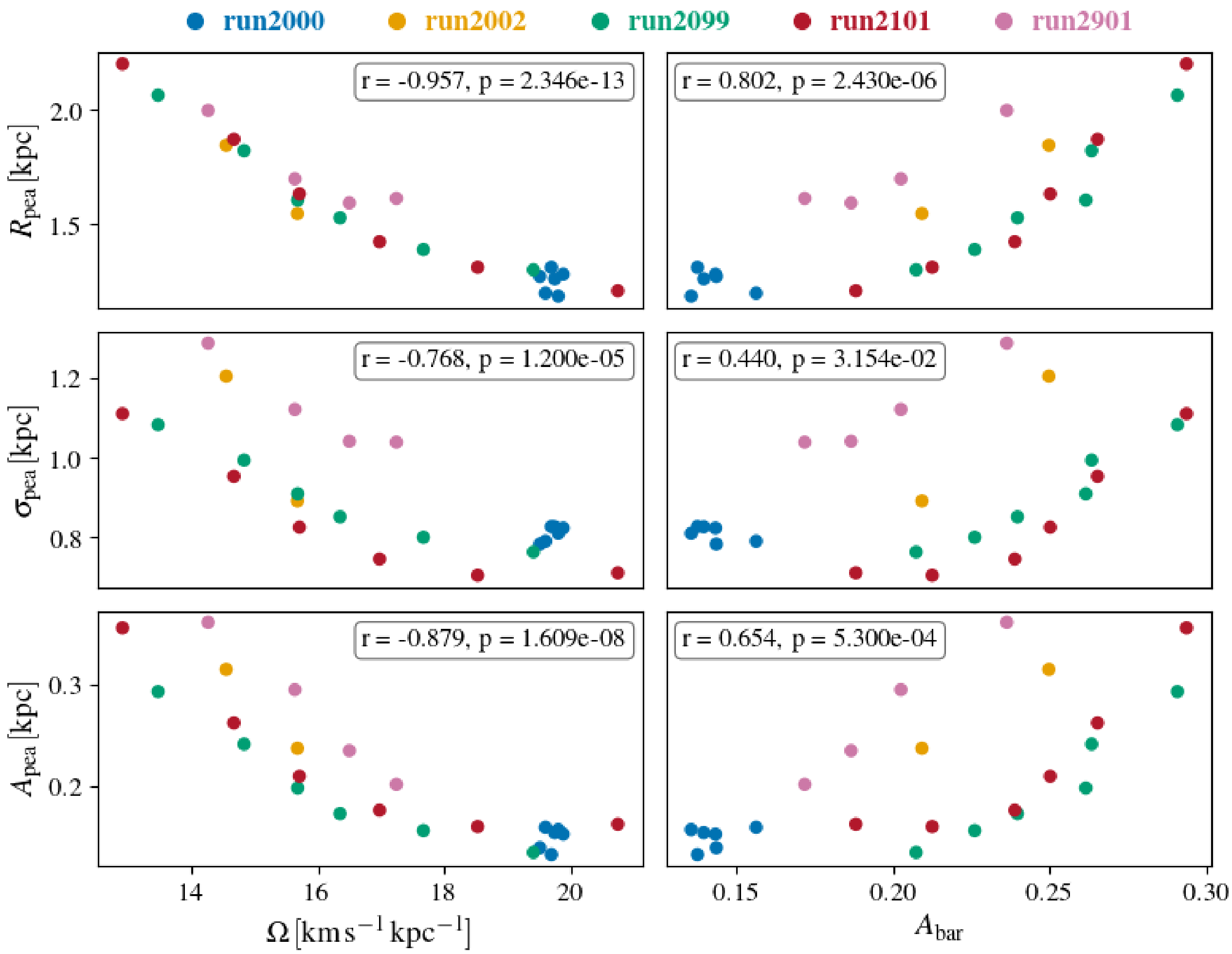

Figure 3. Correlations between BP/X morphology and bar pattern speed and amplutude. Each point is a simulation snapshot; colors identify models *run2000* (blue), *run2002* (orange), *run2099* (green), *run2101* (red), and *run2901* (purple). The left column shows trends versus bar pattern speed Ω, and the right column versus bar amplitude $A_{\rm bar}$. From top to bottom the panels show: (1) the peanut radius $R_{\rm pea}$; (2) the peanut width $\sigma_{\rm pea}$; and (3) the peanut amplitude $A_{\rm pea}$. Pearson correlation coefficients and p–values for each relation are printed in the inset boxes.

the initial guesses are poorly chosen, leading to convergence on erroneous local minima. To mitigate both modeling error (*efficacy*) and excessive runtime (*efficiency*), we adopt a set of constraints. A full and detailed justification of these choices is given in Dattathri et al. (2024); here we summarize the constraints and provide additional discussion of our own findings on improving efficacy and efficiency:

(i) Since we restrict our study to edge-on galaxies, the inclination angle is fixed to $i = 90°$.

(ii) The bar angle is fixed at $\psi = 45°$.

(iii) The position angle of the bar is constrained to $\theta = 0°$.

(iv) Consistent with prior studies that place $c_{\parallel}, c_{\perp} \in [1.5, 5]$ (e.g. Gadotti et al. 2009; Robin et al. 2012; Wheeler et al. 2023; Dattathri et al. 2024), we fix both parameters to $c_{\parallel} = c_{\perp} = 2$.

(v) Because equations (15) and (16) only describe vertically symmetric bars, we limit our photometric fitting to snapshots recorded after bar buckling events. Buckling produces asymmetrical bending out of the galactic plane, as observed in both data (e.g. Raha et al. 1991; Debattista et al. 2004; Martinez-Valpuesta et al. 2006; Łokas 2019; Collier 2020) and simulations (e.g. Erwin & Debattista 2016; Xiang et al. 2021; Cuomo et al. 2023). Fig. 1 shows the timing of these buckling events for each model.

(vi) In equation (16), the "peanut height function" is modified such that the peanut is aligned with the bar's major axis and symmetric about the galactic center.

(vii) In equations (15) and (18), the bar axis ratio is fixed to $q = Y_{\rm bar}/X_{\rm bar} = 0.4$, consistent with both observational constraints (e.g. Rattenbury et al. 2007) and N-body simulation ranges (e.g. Sellwood & Wilkinson 1993; Gadotti et al. 2009).

This set of constraints constitutes the baseline used in Dattathri et al. (2024).

In addition to these, we impose further restrictions on the X-shape parameters to avoid nonphysical solutions. We found that there is no discernible difference between negative and positive values of the X-shape parameters, $R_{\rm pea}$ and $\sigma_{\rm pea}$, in incorrectly converged models. Erroneous fits often featured a negative normalization factor ρ_0 from equation (15). In fact, most erroneous trials were a result of allowing these negative values, leading to the LM algorithm selecting spurious local minima. After this constraint was applied, even implausibly large initial guesses were tested against the LM algorithm, in which it ultimately converged to the correct best-fit model, albeit at a much slower rate than selecting a more accurate initial guess. The algorithm found more success selecting from overestimated values, than underestimated. By constraining all four parameters ($R_{\rm pea}$, $\sigma_{\rm pea}$, $A_{\rm pea}$, and ρ_0) to strictly positive values, we improved both the efficacy and efficiency of the method by Dattathri et al. (2024).

Fig. (2) features two representative simulations, highlighting the N-body model and its unsharp masked image, along with the IMFIT reconstructed result. In *run2000* the BP/X structure stabilizes shortly after formation, whereas in *run2101* it strengthens and becomes more pronounced through the end of the simulation. The remaining models behave similarly to the latter trend, consistent with continued post-buckling growth of the bar amplitude (see Fig. 1). The other three modeled evolutions can be seen in the Appendix.

4. RESULTS AND DISCUSSION

IMFIT was applied every 0.75 Gyr for each model; only post-buckling snapshots are included. Fig. (3) shows statistically significant correlations (all $p \ll 0.05$) between the BP/X diagnostics ($R_{\rm pea}$, $\sigma_{\rm pea}$, $A_{\rm pea}$) and the bar descriptors (pattern speed Ω and amplitude $A_{\rm bar}$). In all cases, the peanut becomes larger and stronger for slower, more massive bars, consistent with the expected inverse relation between bar length and pattern speed and in agreement with the findings of Tahmasebzadeh et al. (2024).

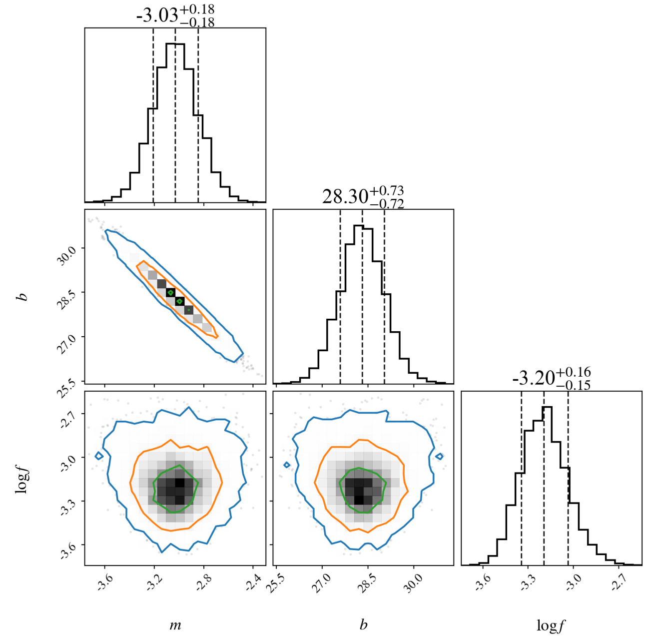

Figure 4. Posterior distributions of the regression parameters describing the correlation between peanut radius and bar pattern speed. The slope (m), intercept (b), and intrinsic fractional-scatter parameter ($\log f$) are shown along the diagonals, together with their median values and 16–84th percentile uncertainties. Contours denote the 68% (green), 95% (orange), and 99.7% (blue) confidence levels.

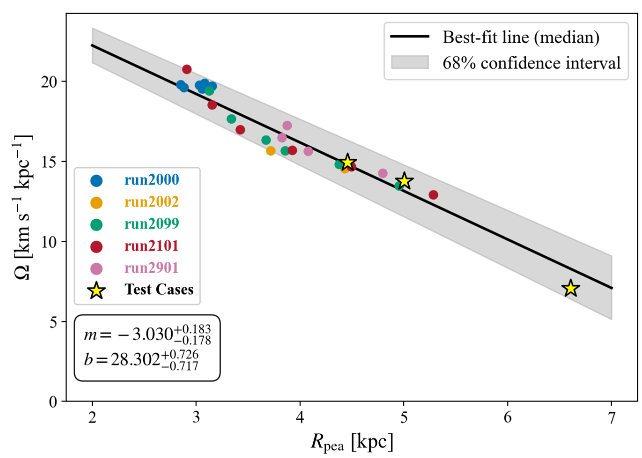

Figure 5. Bayesian linear regression of the relation between peanut length ($R_{\rm pea}$) and bar pattern speed (Ω). Colored circles show post-buckling snapshots for models *run2000* (blue), *run2002* (orange), *run2099* (green), *run2101* (red), and *run2901* (pink). Yellow stars indicate selected test cases. The black line corresponds to the median best-fit relation, with the gray band showing the 68% confidence interval. The inset box gives the slope and intercept with 1σ uncertainties.

Among these trends, the relation between $R_{\rm pea}$ and Ω is the most significant, exhibiting both the lowest scatter and the strongest rejection of the null. Since Ω is measured with uncertainties, this correlation is well-suited for a Bayesian regression framework. We therefore applied MCMC sampling to the marginalized posteriors of the slope, intercept, and intrinsic scatter to quantify the uncertainty in the scaling relation.

In this analysis, we model the correlation between peanut length $R_{\rm pea}$ and bar pattern speed Ω using a linear relation of the form

$$R_{\rm pea} = m\,\Omega + b,$$

where m is the slope, b is the intercept, and we allow for intrinsic scatter following Hogg et al. (2010). The likelihood function is

$$\ln p(y|x,m,b,f) = -\frac{1}{2}\sum_n \left[\frac{(y_n - mx_n - b)^2}{s_n^2} + \ln\left(2\pi s_n^2\right)\right],$$

with $s_n^2 = \sigma_n^2 + f^2\,(mx_n + b)^2$, where σ_n is the measurement uncertainty of y_n and f is a fractional intrinsic-scatter term on y_n. We sample in the parameters $(m, b, \log f)$ because these three quantities fully specify the linear relation and its intrinsic width. This is the standard parameterization for linear regression with intrinsic scatter and is consistent with the recommended formulation of Hogg et al. (2010). Since the total variance entering the likelihood is s_n^2, the posterior uncertainties on (m, b, f) naturally incorporate both the measurement errors and the intrinsic scatter.

We can then infer the relation to estimate Ω for any given galaxy via

$$\Omega = \frac{R_{\rm pea} - b}{m}.$$

Fig. (4) shows the resulting posteriors: the slope and intercept are strongly anti-correlated, reflecting the expected scaling relation, while the scatter parameter is uncorrelated with either, confirming that noise does not bias the regression. The inferred value, $f \simeq e^{-3.2} \approx 0.04$, corresponds to an intrinsic scatter of only $\sim 4\%$ of the model prediction, indicating that the peanut–bar scaling relation is tight. The lack of correlation between $\log f$ and (m, b) confirms that the intrinsic scatter does not drive the slope or intercept and that the regression is not noise-dominated.

The correlations between the BP/X diagnostics and the bar properties reveal that the peanut radius consistently becomes larger for slower, more massive bars, consistent with the expected inverse relation between bar length and pattern speed. The resulting best-fit relation, shown in Fig. (5), confirms that larger peanuts are hosted by slower bars, reinforcing the secular link between bar dynamics and BP/X morphology. The test cases from Wheeler et al. (2023) — specifically *run2950*, *run2952*, and *run6000* — provide an independent validation of the inferred scaling relation. None of these models were included in the original correlation analysis, but all were examined with the method of Dattathri et al. (2024) afterward to assess the applicability of the linear fit. All three fall within the 68% confidence interval of the best-fit relation. In particular, *run6000*, which hosts the largest peanut radius of any model considered in this paper, still lies well within the predicted relation, reinforcing the robustness of the scaling across a broader dynamical range than probed by our primary sample.

A vast amount of evidence demonstrates that the BP/X bulge is not an independent component of the galactic bar, but rather its prominence is correlated to the bar strength and therefore a dependent component of it (Kuijken & Dubinski 1995; Bureau & Freeman 1999; Bureau & Athanassoula 2005; Laurikainen & Salo 2016). Because the bar is closely tied to a wide range of dynamical properties, its connection to the peanut length makes it a valuable indicator of a galaxy's evolutionary history. This link is particularly important in edge-on systems, where little of the bar's structural diagnostics are accessible.

In this study, we deliberately excluded models hosting supermassive black holes, which Wheeler et al. (2023) showed can strongly affect both bar dynamics and BP/X morphology. For end-on bars, the method of Dattathri et al. (2024) faces intrinsic difficulties in recovering the BP/X structure due to its strong dependence on the bar's position angle. Here we constrained the bar orientation to $\psi = 45°$. While the $m = 2$ Fourier mode was adopted to measure bar amplitude, uncertainties on these measurements were not available, preventing the application of a Bayesian MCMC regression analogous to the $R_{\rm pea}$–Ω relation. In principle, $R_{\rm pea}$ may serve as a proxy for $A_{\rm bar}$, but the connection is less compelling than the robust anti-correlation established between $R_{\rm pea}$ and Ω.

Since a majority of disk galaxies host bars, any complete model of an edge-on galaxy must account for their presence and kinematics. In particular, the bar pattern speed is a key driver of secular evolution. The Tremaine–Weinberg method, widely employed in other contexts, fails for edge-on geometries (Dattathri et al.

2024) and can yield errors exceeding 200% outside of a narrow range of orientations (Zou et al. 2019).

We have demonstrated a methodology for estimating the bar pattern speed directly from the peanut radius, providing a new avenue for probing the dynamics of Milky Way–like, edge-on barred galaxies. While our analysis has thus far been applied only to N-body simulations, we are confident the method is readily adaptable to observations of real galaxies.

Finally, we reaffirm that the peanut structure is unambiguously linked to the bar; its size and strength increase with bar amplitude and decrease with pattern speed, mirroring the bar's own scaling relations. Together, these results strengthen the case that the BP/X bulge is not merely associated with the bar, but is instead a fundamental aspect of its three-dimensional structure.

ACKNOWLEDGMENTS

Software: Imfit (Erwin 2015), Matplotlib (Hunter 2007), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020). The additional Imfit modules for modeling the bar can be obtained from the authors.

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APPENDIX

The following figures show additional examples of the other model runs not included in Fig. (2).

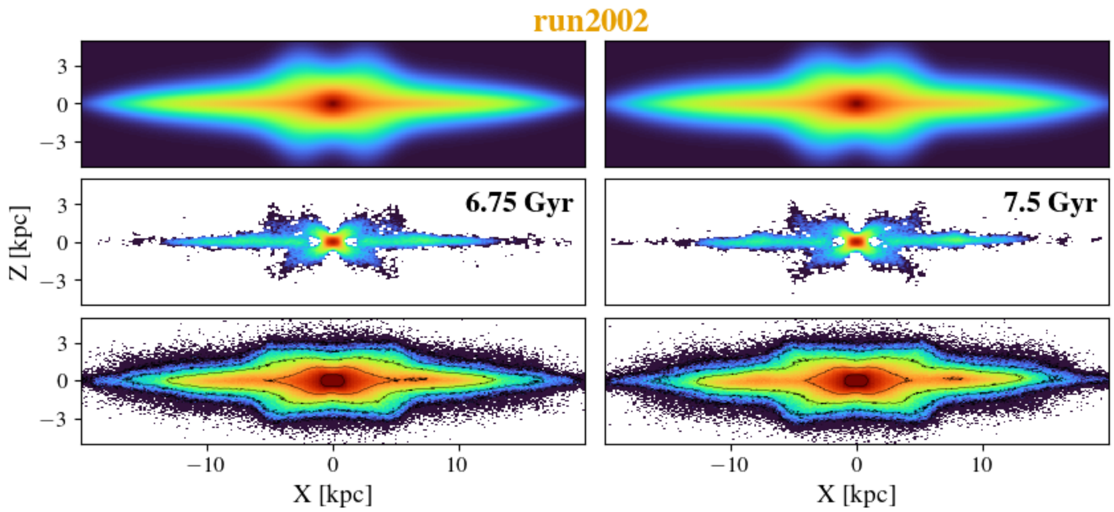

Figure A1. Edge-on structure of model *run2002*. Columns show $t = 6.75$ and 7.5 Gyr, respectively. In each block, from top to bottom the panels display: (1) the IMFIT reconstruction of the edge-on surface density; (2) the unsharp-masked density map of the N-body model, highlighting the BP/X morphology; (3) the surface density of the N-body simulation. The left column snapshot is selected shortly after bar buckling and the formation of the BP/X structure, while the right column is taken at the end of the simulation.

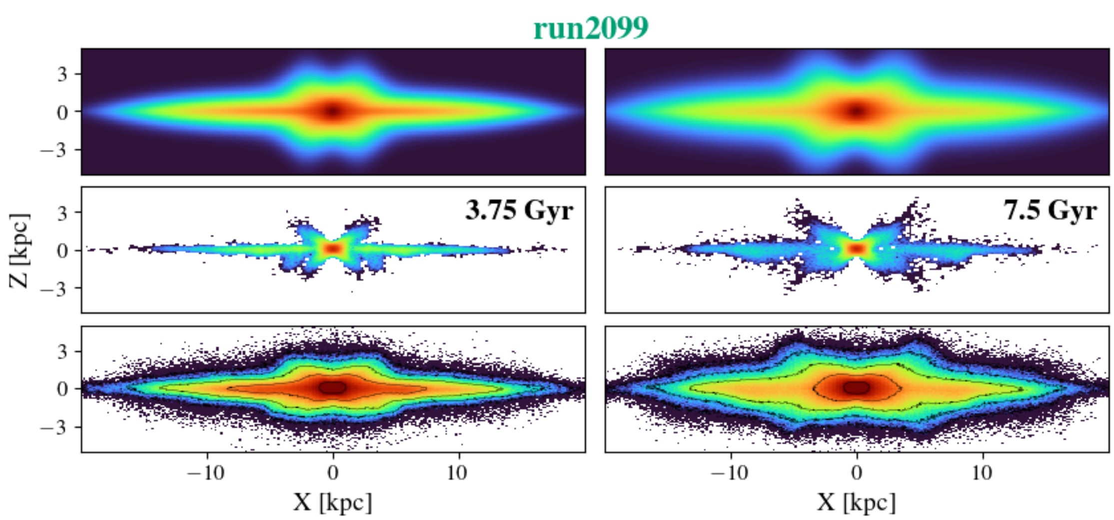

Figure A2. Edge-on structure of model *run2099*. Columns show $t = 6.75$ and 7.5 Gyr, respectively. In each block, from top to bottom the panels display: (1) the IMFIT reconstruction of the edge-on surface density; (2) the unsharp-masked density map of the N-body model, highlighting the BP/X morphology; (3) the surface density of the N-body simulation. The left column snapshot is selected shortly after bar buckling and the formation of the BP/X structure, while the right column is taken at the end of the simulation.

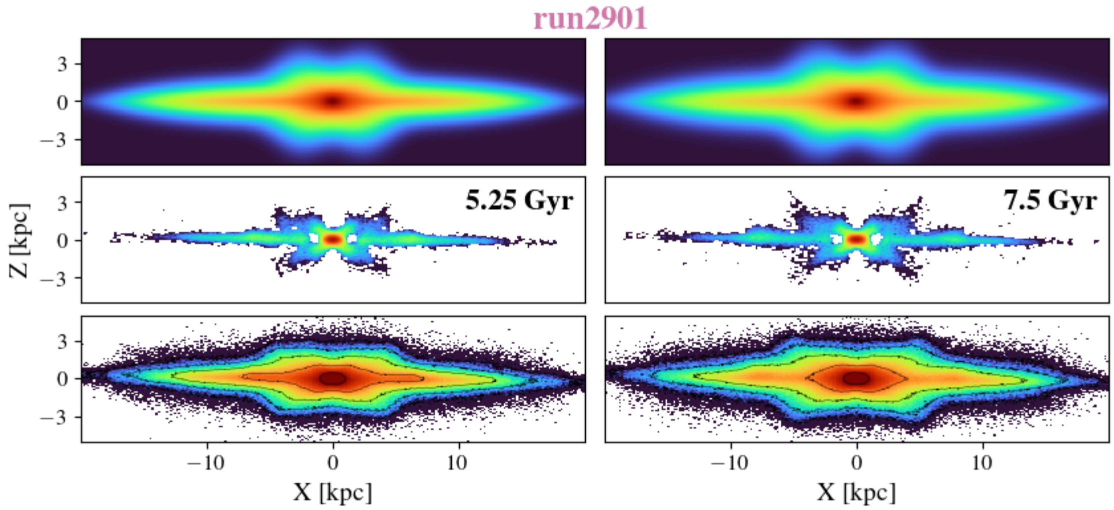

Figure A3. Edge-on structure of model *run2901*. Columns show $t = 5.25$ and 7.5 Gyr, respectively. In each block, from top to bottom the panels display: (1) the IMFIT reconstruction of the edge-on surface density; (2) the unsharp-masked density map of the N-body model, highlighting the BP/X morphology; (3) the surface density of the N-body simulation. The left column snapshot is selected shortly after bar buckling and the formation of the BP/X structure, while the right column is taken at the end of the simulation.